

CHARLES BENELLO

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EDUCATION

The University of Chicago

Chicago, IL

Master of Science in Financial Mathematics

Expected Dec 2026

- **Wayne Booth Graduate Student Teaching Prize** (2026) - UChicago's top graduate teaching award
- Coursework: Options Pricing, Probability & Stochastic Processes, Portfolio & Risk Management, Computing for Finance, ML in Finance, Reinforcement Learning

The University of Chicago | GPA: 3.75/4.0

Chicago, IL

Bachelor of Science in Mathematics and Computer Science (Systems), Minor in Korean

Jun 2025

- Coursework: Algorithms, Database Systems, Distributed Systems, Concurrency Control, Operating Systems, Networks

EXPERIENCE

Quantitative Researcher

Mar 2026 – Present

PayPal (UChicago Project Lab)

Chicago, IL

- Designing **representation strategies** for transaction foundation models by comparing multi-token vs. single-token encoding and time-delta strategies (score-relative vs. inter-event) over **24M+ transactions** for fraud detection.
- Benchmarking **BERT-style masked prediction vs. GPT-style next-token prediction** architectures with ablations across tokenization, time encoding, and context window size.
- Evaluating learned embeddings against traditional ML baselines (**XGBoost, MLP**) on downstream fraud detection with highly imbalanced data (~0.12% positive rate).

Software Engineering Intern

Mar 2026 – Present

Alter Domus

Chicago, IL

- Lifted document extraction accuracy from **50% to 92–97%** by architecting a multi-stage pipeline with schema-enforced structured output and cross-reference verification for 100+ page financial documents (prospectuses, capital calls, regulatory filings).
- Reduced hallucination rates across the platform's AI systems by building an **LLM-as-a-Judge evaluation framework** with automated benchmarks for faithfulness, numerical accuracy, and completeness.
- Cut **LLM inference costs** and improved response speed by optimizing context handling and adding personalization to client-facing portfolio AI.

Quantitative Researcher

Oct – Dec 2025

Bank of America (UChicago Project Lab)

Chicago, IL

- Built an AI-driven platform to **rebalance high-quality liquid asset portfolios**, generating trade recommendations that minimize risk exposure over 6-month projections under capital, liquidity, and net interest income constraints.
- Engineered a **multi-agent LLM debate system** where competing agents generate macro stress scenarios and produce probability-weighted reallocation recommendations; shipped full-stack interfaces for portfolio visualization.

Database Research Assistant

Apr 2023 – Present

ChiData Research Lab, University of Chicago

Chicago, IL

- First-authored **VLDB 2026** paper by designing a parallel external merge sort in **Rust** that completes under tight memory budgets where PostgreSQL, DuckDB, and ClickHouse abort.
- Sole-authored **HPTS 2026** paper by building a lightweight prediction model for sort configuration on cloud/disaggregated hardware, transferring calibration across machines within **3–5% of optimal**.

GPU Systems Researcher

Jun – Sep 2025

EPFL (DIAS Lab)

Lausanne, Switzerland

- Cut out-of-memory query times by **50–200%** by engineering a macroblock-based GPU decompression strategy in **CUDA/C++** and building cost-based performance models for adaptive decompression on TPC-DS workloads.

PROJECTS

Transit Coverage Optimizer | *Python, Combinatorial Optimization* | [cmbenello.github.io/covtsp](https://github.com/cmbenello/covtsp)

2026

- Built an automated Covering TSP solver that beat the Guinness World Record by **60 minutes** (16h45m vs. 17h45m) on a time-expanded graph with 160K nodes and 1.26M edges. Generalized to NYC (475 stations) and Berlin.

Relational Database Engine | *Rust* | github.com/cmbenello/CrustyDB

2023

- Implemented a **multi-threaded relational database from scratch**: B+ tree indexing, concurrent buffer pool management with page-level locking, heap storage, and a query execution engine supporting SQL operations.

TECHNICAL SKILLS

Languages: Python, Rust, C/C++, SQL, CUDA, TypeScript, Go

Frameworks & Tools: PyTorch, XGBoost, NumPy/Pandas, scikit-learn, React, Next.js, Flask, FastAPI, Docker, Git, Linux, PostgreSQL, DuckDB, Databricks

Concepts: Stochastic Processes, Portfolio Optimization, Options Pricing, Machine Learning, Statistical Modeling, Distributed Systems, Database Internals, GPU Programming

PUBLICATIONS

[1] R. Otaki*, **C. Benello***, et al. "CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort." *VLDB 2026*. (*Equal contribution)

[2] **C. Benello**. "Getting the Most Out of External Sorting in Cloud and Disaggregated Environments." *HPTS 2026*.

[3] R. Otaki, J.H. Chang, **C. Benello**, A. Elmore, G. Graefe. "Resource-Adaptive Query Execution." *CIDR 2025*.

ADDITIONAL

Citizenship: United States, United Kingdom, Italy

Languages: Korean, Italian (Intermediate); French (Basic)

Interests: Running (Chicago Marathon '22, '23, '24), Climbing, Weightlifting